

12th National Conference on Earthquake Engineering
Salt Lake City, Utah
27 June - 1 July 2022
Hosted by the Earthquake Engineering Research Institute

Best practices for collecting and using post-earthquake damage data: lessons from Haiti and other past events

S. Loos[1], D. Lallemant[2], and D. J. Wald[3]

ABSTRACT

Researchers often state that "perishable" data need to be collected after a disaster occurs, often referencing the collection of accurate damage information before recovery begins. Damage data, however, are perishable in a different way—they are collected, but they perish due to lack of use during an event. In this study, we review the multitude of damage data sources available—and their potential uses—at different time points after an earthquake occurs. We then ask, have these various data sources been applied to these use cases after past events? Here, we focus on the acquisition and use of damage data after the 2021 Haiti earthquake as compared to previous events, including the 2010 Haiti earthquake. As a community of damage information providers, we find that we are moving in the right direction so that the damage data we acquire can simultaneously serve multiple purposes, ranging from scientific understanding to post-event response and recovery planning. From these events and personal experiences, we close with a set of recommendations and best practices for future post-disaster data collection.

Introduction

Scientists often state that "perishable" data needs to be collected after a disaster, whether related to the hazard itself, scientific understanding, or its societally relevant impacts. For engineers, collection focuses on the acquisition of accurate damage information before repairs begin. Damage data, however, are perishable in a different way—they may be acquired in a timely fashion but perish if used only by academics instead of for local decision makers.

There are both practical and ethical implications of acquiring damage data that is not fully utilized. Practically speaking, many organizations contribute valuable time, energy, and resources towards collecting and analyzing post-earthquake damage information. For example, the USGS devotes several personnel to generate PAGER loss estimates in near real-time after every significant earthquake. Similarly, engineering organizations utilize the time and expertise of multiple professionals to conduct reconnaissance missions, including the Earthquake Engineering Research Institute (EERI), the NSF Structural Extreme Events Reconnaissance (StEER), the Earthquake Engineering Field Investigation Team (EEFIT), or the Geotechnical Extreme Events Reconnaissance (GEER). If these damage data are not utilized, the benefits of labor, development, and production costs are not fully reached.

Disaster researchers also have an ethical responsibility to collect field-surveyed data as part of a coordinated and

[1] Mendenhall Fellow, U.S. Geological Survey, Golden, CO 80401 (email: sloos@usgs.gov)
[2] Assistant Professor, Nanyang Technological University, Singapore, Singapore
[3] Research Seismologist, U.S. Geological Survey, Golden, CO 80401

Loos S, Lallemant D, and Wald DJ. Best practices for collecting and using post-earthquake damage data: lessons from Haiti and other past events. *Proceedings of the 12th National Conference in Earthquake Engineering*, Earthquake Engineering Research Institute, Salt Lake City, UT. 2022.

unified effort with explicit purposes [1]. Coordinated damage data collection mitigates the potential for overwhelming the affected region, with researchers gathering potentially redundant damage statistics. An explicit purpose for data collection mitigates the potential for it to be extractive of affected groups by ensuring the data informs local planning.

This study examines how the damage data acquired after past events have been and might be used. We investigate connections between damage data sources and use cases after the 2021 Haiti earthquake contextualized with past events, including the Italy 2016, Nepal 2015, New Zealand 2011, and Haiti 2010 earthquakes. Here, we provide preliminary results from a systematic review of different sources of damage data and their reported uses from the literature, reconnaissance reports, and media articles. We find post-earthquake damage sources often inform scientific understanding rather than directly informing response, recovery, or mitigation decisions made during each event. The Haiti 2021 earthquake provides an example where damage information was collected and shared intentionally with decision-makers, though future improvements can be made. We therefore close with best practices from these examples to promote coordinated, unified, and useful damage data collection in the future.

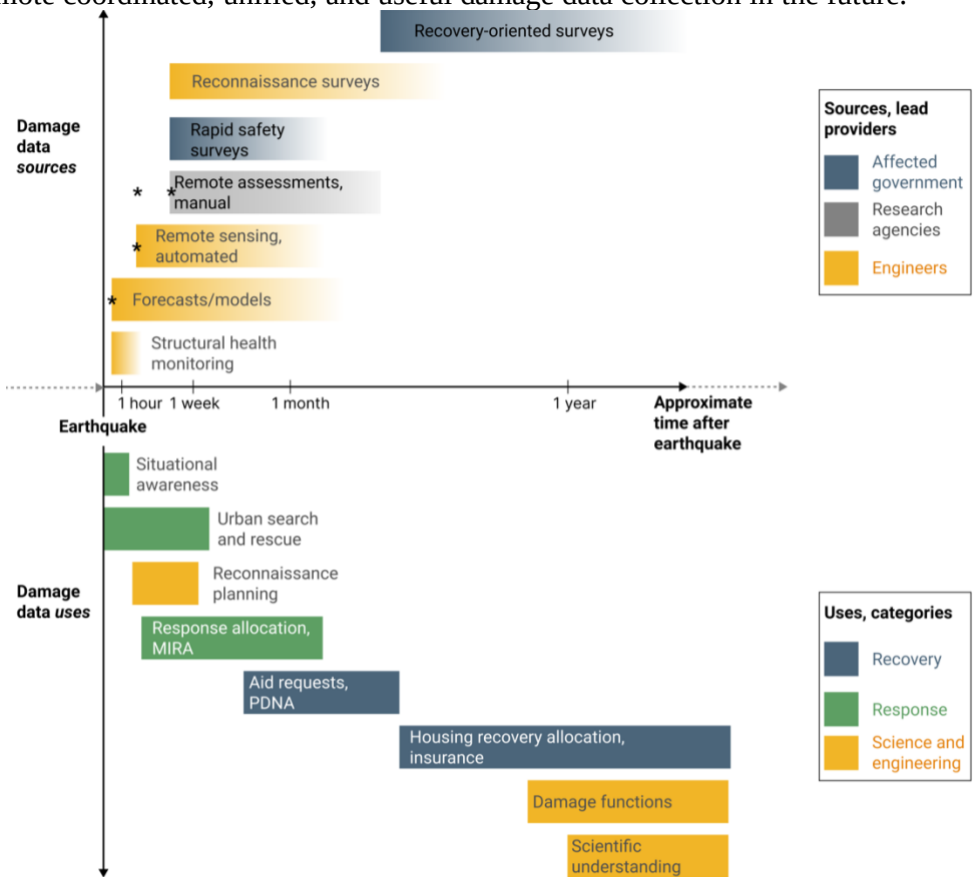


Figure 1. Timeline of availability of sources of damage data (top, modified from [2]), and post-disaster uses of damage data (bottom, modified from [6]). Asterisks (*) indicate data sources for the 2021 Haiti earthquake.

Sources and uses of damage data

Damage data often comes in the form of primary field-surveyed assessments of damage, which are increasingly supplemented by secondary data from rapid forecasts, remote-sensing, remote assessments, and crowdsourcing. The top half of Figure 1 modifies the timeline of the availability of damage data sources from [2] by 1) adding structural health monitoring (SHM) as a new data source and 2) splitting the remote sensing derived datasets into automated and manual assessments. The addition of SHM as a category stems from innovations that increase its prevalence and applicability through systems like Safehub or q-NAVI. These systems allow private building owners to monitor expected damage in a company's building portfolio after an event [3, 4] or Italy's example of using public monitoring data acquired from 150 distributed structures to rapidly evaluate damage states [5]. Splitting the remotely sensed data category stems from the increasing prevalence of virtual reconnaissance teams, such as EERI's Virtual Earthquake Reconnaissance Team (VERT) or StEER's Virtual Assessment Structural Team (VAST) [6, 7], and to acknowledge standardly available manual assessments from Copernicus, UNITAR, and similar agencies [8].

There are also specific uses that depend on these sources of damage information, including response and recovery planning. The bottom half of Figure 1 modifies the timeline of damage information-dependent tasks originally developed in [9], adding uses found in this study or from the authors' experience as developers of damage information.

Comparing the timeline of damage data *sources* and *uses* in Figure 1, certain sources are available in time to inform these uses. For example, rapid forecasts of damage are available in time to inform situational awareness activities, which is often the case for PAGER loss estimates that inform rapid tasks, like disaster declarations [10]. However, we are unaware how often the other damage data sources inform any of the listed tasks in Figure 1.

Connections between damage data sources and uses after past events

The August 14, 2021, Haiti earthquake is one example of an event that resulted in multiple sources of damage data. This event included few reconnaissance teams due to the Covid-19 pandemic and Haiti's security situation. Within 25 minutes, the first damage data source that became available was USGS's PAGER estimate [11]. One day after, NASA-JPL ARIA produced their damage proxy map [12]. On the day of the earthquake, the European Commission Joint Research Center activated their Copernicus Emergency Mapping Service and delivered point estimates of damage manually assessed from optical satellite imagery in the days afterward [13]. Similar remote assessments were also provided by UNITAR [14] and Sertit [15]. Beginning four days after the earthquake, the StEER network collaborated with teams in the Sud Department and GeoHazards International (GHI) in Haiti to collect images of buildings, which were remotely reviewed by skilled earthquake engineers to assign damage ratings [16], thus leading to a primarily virtual reconnaissance (and remote assessments). At the time of this writing, the Government of Haiti is leading a building-level damage survey to plan for recovery [17]. These timeframes are depicted in Figure 1.

Figure 2 shows the connections between damage data sources and uses for the Haiti 2021 earthquake compared to four previous events. Each line represents a use case found in our examination of previous literature and reports or from our experience working on these events. The full list of sources is included in the Appendix.

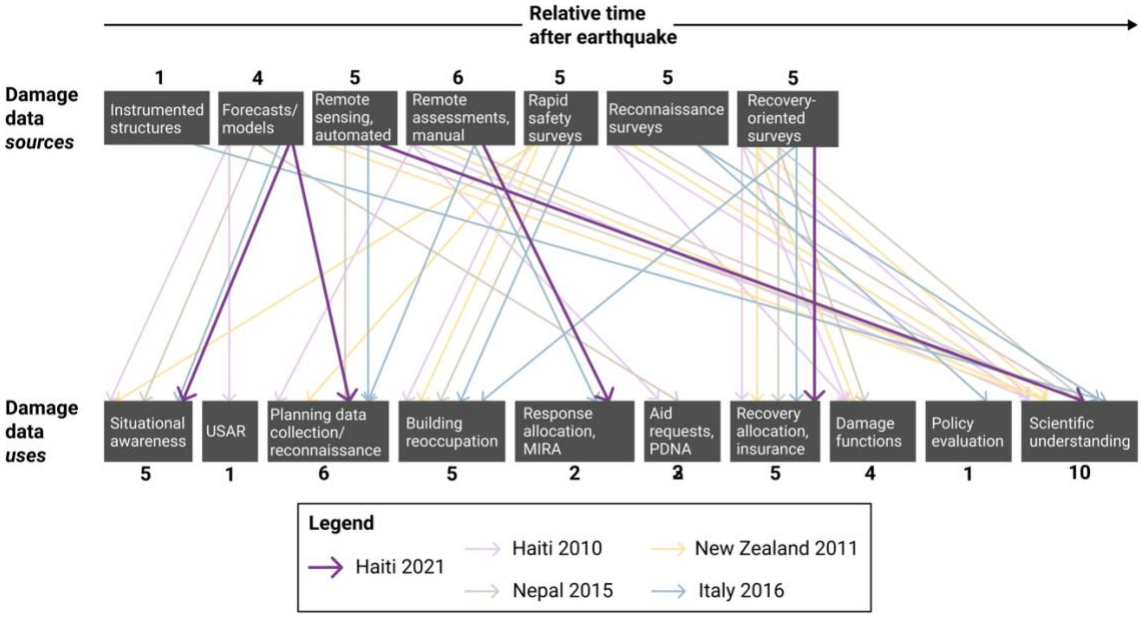


Figure 2. The uses of damage data sources that were acquired after the five earthquakes in this study. The Haiti 2021 data sources and uses are shown in bold purple. Uses are as of December 2021.

Overall, the Haiti 2021 event seems to have more integration between data sources and uses, though documented uses are still being defined as response and recovery unfolds. The USGS PAGER model was used to provide situational awareness to USAID's Bureau of Humanitarian Assistance (BHA) in the days after the event [11]. This is routine since the PAGER system was developed for BHA. The remote assessments from UNITAR, Copernicus, and SERTIT are also standard sources when activated. After the Haiti 2021 event, the UNITAR assessment informed early response allocation [18]. These assessments were also incorporated into StEER's data sharing platform. The Italian government previously used Copernicus assessments after the 2016 earthquakes to allocate response teams and plan reconnaissance surveys (a common use case for remote assessments as shown in Figure 2). StEER's

assessments are an interesting hybrid approach of remote and field surveys, since international engineers remotely assessed pictures taken on the ground by local GHI engineers. This is a potential improvement over past crowdsourcing attempts initiated after the 2010 Haiti earthquake, where engineers were again mobilized to evaluate damage in nadir satellite imagery [19]. Currently, the StEER data has been shared with local NGOs funded by BHA for their shelter projects, and USGS will incorporate these assessments into future updates to shaking estimates, similar to [20]. Finally, the Government of Haiti's recovery-oriented damage surveys will be directly used to inform recovery policies of the affected region [17]. As shown in Figure 2, it is common for recovery surveys to inform recovery aid, because they are designed for that purpose [21]. Importantly, recovery surveys serve multiple purposes–they inform policy and can be used for later scientific studies.

From the analysis presented in Figure 2, there are three main takeaways about the uses of damage data.

1. **Some uses were rarely informed by damage data,** including urban search and rescue, response allocation, and policy evaluation. Italy is a counter-example where Copernicus data were used for the latter two uses [22, 23].
2. **Some damage data consistently informed local uses,** including forecasts/models and recovery-oriented surveys. This is because these damage data sources were designed with a specific purpose in mind.
3. **Many damage data sources only inform scientific understanding** when they could simultaneously inform other local uses by decision-makers. The automated, remote-sensing derived data sources or manual, remote assessments were found to mainly inform reconnaissance planning, whose resulting data then informs damage functions developed *after* the event or reconnaissance articles, instead of tasks that support survivors of the *current* event.

Best practices and recommendations for making damage information useful

Generally, as a research community, we are moving towards making the damage data that we acquire more useful for decision-makers that serve the locally affected population. Based on this preliminary study, we have gathered a few best practices the research community can use to make damage data collection more useful to practitioners.

1. **Openly available data and sharing platforms.** Making data openly available and downloadable by others removes barriers for users to access data. As exhibited by the Haiti 2021 earthquake, StEER took this one step forward and gathered much of the damage data into one hub (https://www.steer.network/haiti-response).
2. **Liaisons between providers and users.** One reason damage data from the Haiti 2021 earthquake is currently being used is due to the existence of liaisons between providers and users of the data. For example, D. J. Wald connected the StEER teams with colleagues at the BHA to ensure StEER damage data could be used for macroseismic assignments for ShakeMap. Similarly, D. Lallemant informed the Government of Haiti of the StEER assessments to plan their recovery-oriented surveys.
3. **Streamlined data collection.** Coordinating who collects data, and where, can mitigate redundant data collection. The Haiti 2021 earthquake was a useful example of a more streamlined approach in that there was one main reconnaissance team (StEER) and one government-led team, though more can be done to connect these two efforts since overlapping information is likely being collected for both. In addition, Copernicus, UNITAR, and Sertit all remotely assessed different regions[18].
4. **Standardized data collection.** Coordinating the method for damage data collection can make it useful for multiple purposes. Often damage data is produced using different scales or units [24], even between survey teams, making it hard to combine. We should also standardize the collection of both damaged and undamaged buildings to be able to constrain shaking estimates [20] and statistical estimates of damage (see next point).
5. **Statistical integration of damage data.** Certain cases require damage estimates at all locations in the affected region before a full field survey can be conducted. In these cases, statistical integration methods can update forecasts with remote sensing and/or field surveyed data to leverage all available data [2, 24, 25].
6. **Standard practice to document uses.** The intended use of damage data should be identified *before* it is acquired, or at the very least, documented afterwards. We found it was not common to document the uses of acquired data. Therefore, a possible use limitation of building damage data is due to this omission bias.

We welcome any feedback or reports of past uses for these and additional events as we continue this research.

Acknowledgments

Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government. We thank Tracy Kijewski-Correa and Janice Rodgers for describing the Haiti StEER data. We also thank Lindsay Davis and Anne Wein for their feedback on this initial draft.

References

[1] Gaillard J, Peek L. Disaster-zone research needs a code of conduct. *Nature* 2019; 575: 440–442.

[2] Loos S, Lallemant D, Baker J, et al. G-DIF: A geospatial data integration framework to rapidly estimate post-earthquake damage. *Earthq Spectra* 2020; 36: 1695–1718.

[3] Safehub, https://safehub.io (accessed 21 December 2021).

[4] Kanda K, Nakashima M, Suzuki Y, et al. "q-NAVI": A case of market-based implementation of structural health monitoring in Japan. *Earthq Spectra* 2021; 37: 160–179.

[5] Dolce M, Di Bucci D. The 2016–2017 Central Apennines Seismic Sequence: Analogies and Differences with Recent Italian Earthquakes. In: Pitilakis K (ed) *Recent Advances in Earthquake Engineering in Europe: 16th European Conference on Earthquake Engineering-Thessaloniki 2018*. Cham: Springer International Publishing, pp. 603–638.

[6] VERT, http://www.learningfromearthquakes.org/activities/vert (accessed 27 December 2021).

[7] Kijewski-correa T, Prevatt DO. Virtual assessment structural team (VAST) handbook : Data enrichment and quality control (DE / QC) for StEER earthquake rapid evaluation form edited by :

[8] Dorati C, Kucera J, Rivero IM i, et al. *Product user manual for copernicus EMS rapid mapping*. JRC111889, Joint Research Center.

[9] Bhattacharjee G, Barns K, Loos S, et al. Developing a user-centric understanding of post-disaster building damage information needs. In: *11th U.S. national conference on earthquake engineering*. Los Angeles, CA, 2018.

[10] Thompson EM, McBride SK, Hayes GP, et al. USGS Near-Real-Time Products—and Their Use—for the 2018 Anchorage Earthquake. *Seismol Res Lett* 2019; 91: 94–113.

[11] Engler D, Jaiswal KS, Noh HY, et al. An accounting of USGS PAGER post-event loss estimates and updates for the 2021 Haiti (M7.2) earthquake. In: *Proceedings of the 12th National Conference in Earthquake Engineering*. Salt Lake City, UT: Earthquake Engineering Research Institute, 2022.

[12] NASA Applied Sciences. Haiti Earthquake, Landslides & Flooding 2021, http://appliedsciences.nasa.gov/what-we-do/disasters/disasters-activations/haiti-earthquake-landslides-flooding-2021 (accessed 22 December 2021).

[13] European Commission, Joint Research Centre (JRC). *Earthquake in Haiti (2021-08-14)*. European Commission, Joint Research Centre (JRC), http://data.europa.eu/89h/e1ea8b62-17cc-4c8d-9406-06fa0848a8dc (2021, accessed 20 August 2021).

[14] UNITAR. Damage Assessment in Nippes Department, Haiti as of 16 August 2021, https://unitar.org/unosat/node/44/3346 (2021, accessed 20 August 2021).

[15] Sertit iCUBE. Charter Call 836-837 (ID 729) - Haiti Earthquake. *ICube-SERTITEU*, https://sertit.unistra.fr/en/rms/ (accessed 19 August 2021).

[16] Kijewski-Correa T, Mosalam K, Robertson I, et al. StEER - 14 August 2021, M7.2 Nippes Earthquake, Haiti. Epub ahead of print 22 September 2021. DOI: 10.17603/h7vg-5691.

[17] 2021 Haiti Earthquake | Shelter Cluster, https://www.sheltercluster.org/response/2021-haiti-earthquake

(accessed 5 January 2022).

[18] United Nations Satellite Centre (UNOSAT). *14 August 2021 M7.2 Haiti Earthquake. Preliminary Satellite-Based Comprehensive Damage Assessment Report*. 3351, Geneva, Switzerland: UNITAR-UNOSAT, https://www.unitar.org/maps/map/3351 (27 August 2021, accessed 5 January 2022).

[19] Corbane C, Saito K, Dell'Oro L, et al. A comprehensive analysis of building damage in the 12 January 2010 Mw7 Haiti earthquake using high-resolution satellite and aerial imagery. *Photogramm Eng Remote Sens* 2011; 77: 997–1009.

[20] McGowan SM, Jaiswal KS, Wald DJ. Using structural damage statistics to derive macroseismic intensity within the Kathmandu valley for the 2015 M7.8 Gorkha, Nepal earthquake. *Tectonophysics* 2017; 714–715: 158–172.

[21] Lallemant D, Soden R, Rubinyi S, et al. Post-Disaster Damage Assessments as Catalysts for Recovery: A Look at Assessments Conducted in the Wake of the 2015 Gorkha, Nepal, Earthquake. *Earthq Spectra* 2017; 33: 435–451.

[22] The European Commission. How the copernicus emergency management service supported responses to major earthquakes in central italy, https://emergency.copernicus.eu/mapping/ems/how-copernicus-emergency-management-service-supported-responses-major-earthquakes-central-italy (2017).

[23] Mazzoni S, Castori G, Galasso C, et al. 2016–2017 Central Italy Earthquake Sequence: Seismic Retrofit Policy and Effectiveness. *Earthq Spectra* 2018; 34: 1671–1691.

[24] Loos S, Levitt J, Tomozawa K, et al. Efficacy of damage data integration: a comparative analysis of four major earthquakes. *Rev*.

[25] Xu S, Dimasaka J, Wald DJ, et al. Seismic Multihazard Estimation via Causal Inference from Satellite Imagery. *Rev*.

[26] Lemoine G, Corbane C, Louvrier C, et al. Intercomparison and validation of building damage assessments based on post-Haiti 2010 earthquake imagery using multi-source reference data. *Nat Hazards Earth Syst Sci Discuss* 2013; 1: 1445–1486.

[27] Ghosh S, Huyck CK, Greene M, et al. Crowdsourcing for Rapid Damage Assessment: The Global Earth Observation Catastrophe Assessment Network (GEO-CAN). *Earthq Spectra* 2011; 27: 179–198.

[28] Voigt S, Schneiderhan T, Twele A, et al. Rapid Damage Assessment and Situation Mapping: Learning from the 2010 Haiti Earthquake. *Photogramm Eng Remote Sens* 2011; 77: 923–931.

[29] Government of the Republic of Haiti. *Haiti Earthquake PDNA: Assessment of damage, losses general and sectoral needs*. Port-au-Prince, Haiti.

[30] de Ville de Goyet C, Sarmiento JP, Grünewald F. *Health response to the earthquake in Haiti: Lessons to be learned for the next massive sudden-onset disaster*. Washington, DC: Pan American Health Organization, https://www.paho.org/en/documents/health-response-earthquake-haiti (2011, accessed 5 January 2022).

[31] Eberhard M, Baldridge S, Marshall J, et al. *The MW 7.0 Haiti Earthquake of January 12, 2010: USGS/EERI Advance Reconnaissance Team Report*. Open File Report, https://pubs.usgs.gov/of/2010/1048/ (accessed 20 December 2021).

[32] DesRoches R, Comerio M, Eberhard M, et al. Overview of the 2010 Haiti Earthquake. *Earthq Spectra*

2011; 27: 1–21.

[33] Molina S, Torres Y, Benito B, et al. Using the damage from 2010 Haiti earthquake for calibrating vulnerability models of typical structures in Port-au-Prince (Haiti). *Bull Earthq Eng*; 12. Epub ahead of print 1 August 2013. DOI: 10.1007/s10518-013-9563-z.

[34] Lallemant D, Kiremidjian A. A Beta Distribution Model for Characterizing Earthquake Damage State Distribution. *Earthq Spectra* 2015; 31: 1337–1352.

[35] World Bank. *What Did We Learn? The Shelter Response and Housing Recovery in the First Two Years after the 2010 Haiti Earthquake*. Washington, DC: World Bank, https://openknowledge.worldbank.org/handle/10986/26729 (2016, accessed 5 January 2022).

[36] Foulser-Piggott R, Spence R, Eguchi R, et al. Using Remote Sensing for Building Damage Assessment: GEOCAN Study and Validation for 2011 Christchurch Earthquake. *Earthq Spectra* 2016; 32: 611–631.

[37] Galloway B, Hare J, Brunsdon D, et al. Lessons from the Post-Earthquake Evaluation of Damaged Buildings in Christchurch. *Earthq Spectra* 2014; 30: 451–474.

[38] Marshall JD, Jaiswal K, Gould N, et al. Post-Earthquake Building Safety Inspection: Lessons from the Canterbury, New Zealand, Earthquakes. *Earthq Spectra* 2013; 29: 1091–1107.

[39] Earthquake Engineering Research Institute. *Learning from earthquakes: The M6.3 Christchurch, New Zealand, Earthquake of February 22, 2011*. May, Oakland, CA, https://www.eeri.org/site/images/eeri_newsletter/2011_pdf/EERI_NewZealand_EQRpt_web.pdf (2011).

[40] Van Ballegooy S, Malan P, Lacrosse V, et al. Assessment of liquefaction-induced land damage for residential Christchurch. *Earthq Spectra* 2014; 30: 31–55.

[41] Horspool NA, King AB, Lin SL, et al. Damage and losses to residential buildings during the Canterbury Earthquake Sequence. In: *Proceedings of the New Zealand Society for Earthquake Engineering 2016 Conference*. New Zealand Society for Earthquake Engineering, http://www.nzsee.org.nz/db/2016/Papers/O-04 Horspool.pdf (2016).

[42] Government of Nepal National Planning Commission. *Post Disaster Needs Assessment, Nepal Earthquake 2015*. B, Kathmandu: National Planning Commission, 2015.

[43] Lizundia B, Shrestha SN, Bevington J, et al. *EERI Earthquake Reconnaissance Team Report: M7.8 Gorkha, Nepal Earthquake on April 25, 2015 and its aftershocks*. May.

[44] Government of Nepal National Reconstruction Authority. *Nepal earthquake 2015: Post Disaster Recovery Framework 2016-2020*. Kathmandu, Nepal: National Reconstruction Authority.

[45] Nguyen M, Lallemant D. Order Matters: The Benefits of Ordinal Fragility Curves for Damage and Loss Estimation. *Risk Anal*; n/a. DOI: 10.1111/risa.13815.

[46] Wilson B. Evaluating the INLA-SPDE approach for Bayesian modeling of earthquake damages from geolocated cluster data. Epub ahead of print 2020. DOI: 10.31223/OSF.IO/64WHM.

[47] Sextos A, De Risi R, Pagliaroli A, et al. Local site effects and incremental damage of buildings during the 2016 Central Italy Earthquake sequence. *Earthq Spectra* 2018; 34: 1639–1669.

[48] D'Ayala D, Faure-Walker J, Mildon Z, et al. *The mw 6.2 amatrice, italy earthquake of 24th august 2016, a*

field report by EEFIT. May, EEFIT, 2019.

Appendix

Table 1. Sources for use cases shown in Figure 2

Event	Data Source	Data Use	Citation
Haiti 2010	Forecast/model	Situational awareness	D. J. Wald (personal communication, January 12, 2010)
Haiti 2010	Forecast/model	USAR activation	D. J. Wald (personal communication, January 12, 2010)
Haiti 2010	Crowdsourced or manually derived	Scientific understanding	[19, 26–28]
Haiti 2010	Crowdsourced or manually derived	Reconnaissance planning	[29]
Haiti 2010	Crowdsourced or manually derived	PDNA	[29]
Haiti 2010	Rapid safety surveys	Building reoccupation	[30]
Haiti 2010	Reconnaissance	Scientific understanding	[31, 32]
Haiti 2010	Reconnaissance	Damage functions	[33]
Haiti 2010	Recovery-oriented surveys	Scientific understanding	[34]
Haiti 2010	Recovery-oriented surveys	Recovery aid	[35]
Haiti 2010	Recovery-oriented surveys	Damage functions	[34]
New Zealand Feb 2011	Forecast/model	Situational awareness	D. J. Wald (personal communication, January 12, 2010)
New Zealand Feb 2011	Remote sensing, automatically derived	Scientific understanding	S. Loos (personal communication, September 2018)
New Zealand Feb 2011	Crowdsourced or manually derived	Scientific understanding	[36]

New Zealand Feb 2011	Rapid safety surveys (windshield surveys)	Situational awareness	[37]
New Zealand Feb 2011	Rapid safety surveys	Building reoccupation	[37, 38]
New Zealand Feb 2011	Rapid safety surveys (indicator buildings)	Planning data collection (re-inspection after aftershocks)	[37]
New Zealand Feb 2011	Reconnaissance surveys	Scientific understanding	[39]
New Zealand Feb 2011	Recovery-oriented surveys	Recovery aid	[40]
New Zealand Feb 2011	Recovery-oriented surveys	Scientific understanding	[40]
New Zealand Feb 2011	Recovery-oriented surveys	Damage functions	[41]
Nepal 2015	Modeled/forecasts	Situational awareness	[21]
Nepal 2015	Modeled/forecasts	Aid requests/PDNA	[42]
Nepal 2015	Remote sensing, automatically derived	Scientific understanding	[21]
Nepal 2015	Crowdsourced or manually derived	Scientific understanding	[21]
Nepal 2015	Rapid safety surveys	Building reoccupation	[21]
Nepal 2015	Reconnaissance surveys	Scientific understanding	[43]
Nepal 2015	Recovery-oriented surveys	Recovery policy	[21, 44]
Nepal 2015	Recovery-oriented surveys	Damage functions	[45]
Nepal 2015	Recovery-oriented surveys	Scientific understanding	[2, 46]
Italy 2016	Modeled/forecasts	Situational awareness	[5] D. J. Wald (personal communication, Aug 2016)

Italy 2016	Remote sensing, automatically derived	Planning reconnaissance	[47]
Italy 2016	Crowdsourced or manually derived	Planning reconnaissance	[22]
Italy 2016	Crowdsourced or manually derived	Response allocation (temp shelter)	[22]
Italy 2016	Rapid safety surveys	Building reoccupation	[22]
Italy 2016	Reconnaissance surveys	Scientific understanding	[47, 48]
Italy 2016	Reconnaissance surveys	Damage functions (expected)	[47]
Italy 2016	Reconnaissance surveys	Policy evaluation	[23]
Italy 2016	Recovery-oriented surveys (damage and usability)	Building reoccupation	[5]
Italy 2016	Recovery-oriented surveys (damage and usability)	Reconstruction grant	[5]
Haiti 2021	Forecasts/models	Situational awareness	[11]
Haiti 2021	Forecasts/models	Planning reconnaissance	[16]
Haiti 2021	Remote assessments (manual	Response allocation	[18]
Haiti 2021	Recovery-oriented surveys	Recovery policy	[17]